

## Oct 1 — 5 papers

Episode 18 · October 1, 2026 · 5 papers

**Cascaded consensus splitting for multi-branch contingency games**

SKIM

Bastien Lechardoy et al.

*They extend contingency games so each agent's intent can be revealed at its own branching time, not all at once, and split the resulting game by scenario with ADMM so the branches solve in parallel.*

**PROBLEM** Existing contingency games branch at a single time, which assumes every agent's intent becomes clear at the same moment, and that is wrong when agents reveal their plans at different times. Solving all scenario-dependent best responses jointly also becomes expensive quickly as the number of agents and intents grows.

**RESULTS** The abstract gives no numbers. It says only that, on tightly coupled three-agent interactions, the multi-branch version beats the single-branch baseline and has a lower mean receding-horizon solve time. The wording is 'support that', which is hedged, and the abstract names no benchmark, no success or safety metric, and no other baselines.

**METHOD** They build a tree of trajectories in which one source of intent uncertainty is resolved at each branching time. This gives a shared prefix, then branches, then further branches, so the plan can adapt as each agent's intent becomes clear. To solve it, they give each scenario its own copy of the variables, add consensus constraints that force shared prefixes to agree, and run ADMM so the scenario subproblems can be solved in parallel.

**CAVEAT** The whole evaluation is three agents, so the scaling argument that motivates the solver is never tested at the sizes where it would matter. A tree with several branching levels grows exponentially with agents and intents. ADMM consensus often converges slowly and loosely, and the abstract says nothing about iteration counts, tolerance, or what happens when the solver stops early inside a receding-horizon loop. Branching times also look fixed in advance rather than triggered by when an intent is actually observed, so the claim of adapting to asynchronous revelation may depend on a hand-set schedule.

**WHY IT MATTERS** This sits at the edge of your work. It is interaction-aware planning under intent uncertainty that the planner has to infer, which is exactly the no-communication setting your research tries to replace. Two things carry over. First, the tree with staggered branching times is a natural model for a negotiation where each vehicle's acceptance or intent message arrives at a different time or not at all, so you could reuse it to plan through partial participation or delayed replies. Second, the ADMM consensus splitting is a standard tool you may already use for distributed cooperative planning. It is worth a skim for the branching structure, but it contains no protocol, message, or right-of-way resolution content.

arXiv:2609.37612v1 PDF incremental

**Agent-Based Evolutionary Dynamics for Mixed Autonomy Weaving Ramps**

SKIM

Sheryl Paul et al.

*They take an existing Wardrop-style model of weaving ramps with altruistic automated vehicles and rebuild it as a population of agents that each adjust their lane choice with an evolutionary game rule, then show the agents settle at the same equilibrium the macroscopic model predicts.*

**PROBLEM** Earlier weaving-ramp models showed that altruistic connected vehicles can improve traffic at the population level. They did not show how that equilibrium comes about from individual vehicles acting on their own, or how it holds up with finite fleets, mixed levels of altruism and noisy state information.

**RESULTS** The abstract reports close agreement with the macroscopic predictions but gives no numbers. It names no efficiency gains, convergence times or error magnitudes, and it names no baseline beyond the macroscopic model itself. From the abstract we only know qualitatively that heterogeneity and imperfect information change both system performance and how the altruistic burden is split across vehicles.

**METHOD** Each vehicle picks a lane or route through the weaving section and revises that choice with an evolutionary game-theoretic update. Its payoff mixes its own travel cost with an altruism weight on the cost it imposes on others. They prove that these decentralized dynamics converge to the unique equilibrium of the original macroscopic model. They then simulate the agents to study how fast convergence happens, how the system tracks changing demand, what happens when altruism levels differ between vehicles, and how noisy state information changes the outcome.

**CAVEAT** The vehicles never negotiate. Coordination comes from repeated best-response style lane choice against an aggregate traffic state, so it says nothing about how a specific lane change is agreed between two vehicles, or what happens when a message is lost. The convergence proof is for an evolutionary update toward a known unique Wardrop equilibrium, which is close to a textbook result for potential-like congestion games, so the proof itself is likely a modest step. Evaluation is simulation only, in a six-page workshop paper built on the authors' own earlier macroscopic model, with no microscopic traffic simulator and no comparison to explicit coordination schemes.

**WHY IT MATTERS** This sits mostly outside your work. It is about population-level lane choice driven by altruistic incentives, not about vehicles exchanging intentions or trajectories to agree on a merge. The one link is the imperfect-information result and the question of who carries the altruistic burden. That could give you a fairness or incentive argument for why vehicles would accept a yielding proposal in a negotiation protocol.

arXiv:2609.36424v1 PDF incremental

## Game-Theoretic Control with Constrained Potential Surgery

Zhiyuan Zhang, Panagiotis Tsiotras

SKIM

*A faster interior point solver for constrained dynamic games, plus a cheap second-order fix that makes it less likely to land on saddle points that are not Nash equilibria, tested on scaled race cars.*

**PROBLEM** Newton-type solvers for generalized Nash equilibria only enforce first-order conditions, so they can converge to saddle points where some agent would still want to deviate. They are also too slow to handle more than about four agents in real time.

**RESULTS** The abstract gives no numbers. It claims evaluation on numerical benchmarks and one physical experiment with scaled race cars, but it names no baseline solver, no agent count, no solve times, and no rate of convergence to a true equilibrium.

**METHOD** They build an interior point solver for constrained general-sum dynamic games. On top of it they add a second-order correction, which the title calls potential surgery. It appears to modify the local problem so that the solver is pushed toward points that satisfy each player's second-order optimality conditions, not just stationarity. The abstract does not say whether the correction comes with guarantees or is only a heuristic that raises the odds of success.

**CAVEAT** The central claim is only that the method increases the probability of reaching a local equilibrium, so the paper stands or falls on a head-to-head comparison against existing game solvers such as ALGAMES or the PATH-based mixed complementarity solvers, at matched agent counts and on matched problems, and the abstract shows no such comparison. The abstract also names the four-agent barrier as the motivation but never says how many agents they actually reach. And a scaled race car demo is a narrow, low-stakes test that does not show robustness across different interaction types.

**WHY IT MATTERS** This is only partly relevant to you. It gives a centralized equilibrium solver that implicitly settles who yields between vehicles with conflicting plans, which touches your conflict resolution thread. But it involves no message exchange, no proposal and acceptance, and no handling of lost messages or of vehicles that do not participate. It helps if you use a game-theoretic solve as the back end of a negotiation protocol, or as an upper-bound baseline for what coordination could achieve, and the saddle point issue is a real pitfall to know about if you do.

arXiv:2609.31901v1 PDF incremental

## GlowTact: Simple and Compact Vision-Based Tactile Sensing with High Sensitivity and Spatial Resolution

Yuxiang Ma et al.

SKIM

*GlowTact is a fingertip tactile sensor where a pressed nitrile membrane glows brighter where the pressure is higher, so the camera image is the pressure map with no geometry reconstruction step, and it has nothing to do with connected vehicles.*

**PROBLEM** Most vision-based tactile sensors need multi-color or directional lighting plus a reconstruction step to recover contact geometry, and that makes them hard to shrink to humanoid fingertip size. The authors want an optical principle simple enough to fit small, custom fingertip shapes without giving up sensitivity or spatial detail.

**RESULTS** The abstract says the sensor detects contacts at the scale of grams, estimates normal force accurately, and resolves fine geometry such as the threads of an M1 screw. It gives no force error figures, no spatial resolution in millimetres and no comparison against GelSight-style baselines, so the headline claims can't be checked from the abstract.

**WHY IT MATTERS** This is off-topic for your work. It is hardware for robot hands, with no link to vehicle negotiation, coordination protocols, platooning, conflict resolution or cooperative trajectory planning. It probably reached your feed through the broad robotics category, which is worth tightening in your topic filter.

arXiv:2609.32471v1 PDF notable

**METHOD** They build a membrane that responds to pressure optically, so a single color of diffuse light makes a raw camera image whose brightness tracks local contact pressure. Because of this they need no photometric stereo and no depth estimation, and the optics and packaging stay minimal. The contact skin is a nitrile membrane, which they present as more durable than the usual soft elastomer surfaces.

**CAVEAT** If image intensity stands in for pressure, the reading could be confounded by membrane ageing, uneven lighting, hysteresis and temperature, and the abstract reports no drift, repeatability or long-term durability tests to back the durability claim. Shear and tangential force are not mentioned, and a pressure-only image may lose the slip and shear cues that GelSight-type sensors capture through marker tracking.

Albert Škegro et al.

*A battery-pack controller that ranks the cells at every step and only lets the solver switch on the top few, so the on-off decision grows linearly with pack size instead of exploding combinatorially.*

**PROBLEM** Mixed-integer model predictive control of packs where each cell can be bypassed has to choose among every possible subset of engaged cells, which is too slow for real-time use as packs grow. Exact solvers branch too much, and simple rule-based switching gives up the benefits of optimization.

**RESULTS** In simulation, a 20-cell pack driven over the WLTP light-vehicle test cycle shows several gains over the full subset-selection controller. Every active step is certified, no instance needs branching, 95th-percentile solve time drops by a factor of 6.8, and the cycle-mean spread in state of charge between cells falls by a factor of 3.4 with no power curtailment. Same-state reference solves put the mean restriction gap below 0.11 percent on steps where both controllers certify.

**WHY IT MATTERS** This is peripheral to your work. It is about battery cell switching, with no vehicles, no communication and no negotiation. The one idea you could borrow is general: when a mixed-integer planner has to choose among combinatorially many discrete options, rank the candidates with a cheap score and let the solver pick only a prefix of the ranking. That could apply to choosing merge order or platoon insertion position, but you would need to supply the ranking and the justification yourself.

arXiv:2609.32032v1 PDF incremental

**METHOD** At each sampling instant, every cell gets a one-step suitability score based on the control objective. The engagement variables are then forced to follow that ranking, so the only allowed choice is how many of the top-ranked cells to engage. The model, cost and physical constraints stay the same. Only the feasible set shrinks, from all subsets down to a number of options equal to the cell count.

**CAVEAT** The 0.11 percent gap is only measured on steps that both controllers certify. That leaves out exactly the hard steps where the full controller struggles, which is where a greedy one-step ranking is most likely to be wrong. The 3.4-fold balancing gain probably reflects the full controller running out of time and returning worse solutions, not the restricted problem being better. So it is a real-time budget artifact, not evidence that the restriction is harmless. All results come from simulation of one drive cycle, with no hardware-in-the-loop test and no proof of when a prefix-ordered solution can be optimal.